

IAM Brief

Provisioning Infrastructure: Deprovisioning Gap Analysis

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Summary

A cross-reference of the dstools provisioning scripts against their active crontab schedules reveals that **all deprovisioning and offboarding automation is effectively inactive**. Duo account deletion, AD offboarding, and IDTREE role purge scripts either have their destructive operations commented out or were never added to the crontab. The evidence points to an incomplete Perl-to-Python migration circa 2020–2022: replacement scripts were written but never scheduled, and the original Perl scripts were archived without successors taking their place.

Findings

Duo Account Offboarding — Disabled

`duo_offboard_users.py` is scheduled in the duo crontab (daily at 5:40a). However, the critical API call `admin_api.delete_user()` is commented out on line 294. The script runs daily, builds a list of terminated, retired, and unaffiliated users, writes that list to `delete-disable-users-duo.json`, and then exits without taking action. A header comment in the script notes it was “originally a clone of the sync user attributes script,” suggesting it was adapted from another script and never completed. A hardcoded user skip (`cam0794`) further indicates the script was frozen mid-testing.

A second script, `duo_students_old_enrollment_no_affiliation.py`, targets disabled/terminated student accounts for deletion. It is not scheduled in any crontab, and its `delete_user()` call is also commented out. No Duo scripts in the current infrastructure perform any account deletion or status changes.

IDTREE Role Purge Scripts — Never Scheduled

Three Python purge scripts exist in the repository: `purge_unt_roles.py`, `purge_hsc_roles.py`, and `purge_da1_roles.py`. None appear in the purge crontab. The only scheduled purge job is `find_old_enrollment_groups.py` (weekly, Saturdays). The `purge/archive/` directory contains the original Perl predecessors (`purge_untst_roles.pl`, `purge_hscst_roles.pl`, etc.), confirming that the Python scripts were intended as replacements. The result is approximately 113,000 users retaining stale student attributes—including `classEnrollment` references, role values, and scoped affiliations—indefinitely in IDTREE.

AD Offboarding — Never Scheduled

`ad/offboarding.py` contains logic to disable AD accounts and delete Exchange ActiveSync objects, but it does not appear in any crontab. The only scheduled process that performs any offboarding-adjacent work is `idtree/idtree_offboarding_workaround.py` (daily at 11:59p), which removes terminated employees from IDTREE group memberships—including the DuoHelpDesk group—but does not disable accounts or touch AD or Duo.

Root Cause Assessment

The pattern is consistent across all three areas: working destructive scripts exist in the repository but were never activated. The most probable cause is an incomplete Perl-to-Python migration undertaken between 2020 and 2022. The presence of archived Perl originals alongside unscheduled Python replacements, combined with commented-out API calls and hardcoded test-user skips, indicates that the developer(s) completed the identification/logic phase of the rewrite but deferred enabling the destructive operations—likely out of appropriate caution—and then the project stalled before the final activation step. No documentation or handoff records exist for this work.

Downstream Impact

- **Duo:** Terminated, retired, and unaffiliated users retain active Duo accounts indefinitely, inflating the license count and expanding the authentication surface.
- **IDTREE:** ~113,000 users carry stale enrollment, role, and affiliation attributes, contributing to group bloat (e.g., UNT Graduate Students) and inaccurate affiliation-based access decisions.
- **AD:** No automated account disablement for separated employees via the provisioning server; any offboarding that occurs depends on manual action or Identity Manager drivers.

Remediation Status

The IAM team's current refactoring effort—migrating scripts from `dstools` to the `iam-scripts-provisioning` repository under the `iam_modules` shared library (v3.0.0)—addresses this gap directly. All refactored scripts follow a dry-run-by-default pattern (`--commit` flag required for writes), use modern `ldap3` bindings, and are governed by a formal `CONTRACT.md`. The `iam-scripts-alma-v2` deactivation/reactivation scripts already integrate with Duo, AD, eDirectory, and Entra ID. Completing the migration and scheduling the new purge and offboarding scripts will close the deprovisioning gap that has been open since approximately 2022.